Functions, Rules, Domain & Range

ROUND 1 OF 3 Questions 1–4

Work independently, then compare your reasoning with your group.

Question 1 R&P

A function machine follows $y = 2x + 5$. Complete the outputs for inputs 0, 2, 4, and 6.

Question 2 R&P

For the relation {(-1,0), (0,2), (-1,3), (2,8)}, decide whether it is a function. Explain and list domain/range.

Question 3 R&P

Write two different ordered pairs with the same OUTPUT but different INPUTS. Does this break the function rule?

Question 4 R&P

A game costs $5 to enter, plus $3 per round. Use inputs 0, 1, 2, 3 rounds to list the outputs and the domain/range.

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ROUND 2 OF 3 Questions 5–8

Work independently, then compare your reasoning with your group.

Question 5 **R&P**

Spiral: Solve $2(x + 5) = 24$. Show your steps.

Question 6 **R&P**

Preview: Table outputs are 9, 6, 3, 0 for inputs 1, 2, 3, 4. What stays constant as the input increases by 1?

Question 7 **Matching practice**

Use the rule y = 3x - 2. Find the outputs for inputs 0, 1, 3, and 5.

Question 8 **Matching practice**

For {(-2,5), (0,1), (-2,5), (3,7)}, decide whether the relation is a function and list its domain and range.

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ROUND 3 OF 3

Questions 9–12

Work independently, then compare your reasoning with your group.

Question 9 — **Matching practice**

Write two different ordered pairs with the same output of 6, but different inputs. Does that violate the function rule?

Question 10 — **Matching practice**

A bike rental costs $4 plus $2 per hour. List costs for 0, 1, 2, and 4 hours, including domain and range.

Question 11 — **Matching practice**

Solve 3(x - 2) + 4 = 22. Show your steps.

Question 12 — **Matching practice**

A table gives outputs 12, 8, 4, 0 for inputs 0, 1, 2, 3. What is the constant output change per +1 input?